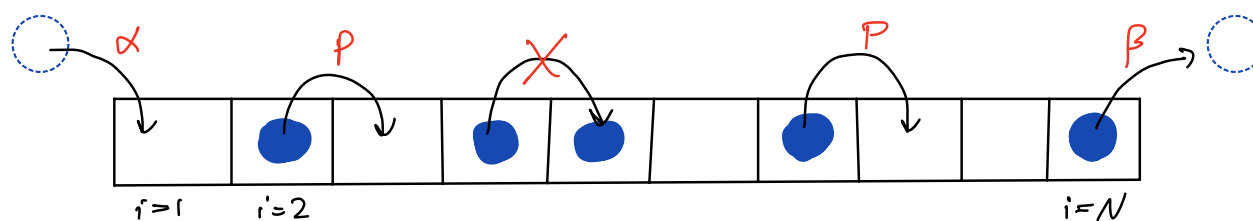


## Recitation 9 - The Totally Asymmetric Simple Exclusion Process (TASEP)

### Model definition:

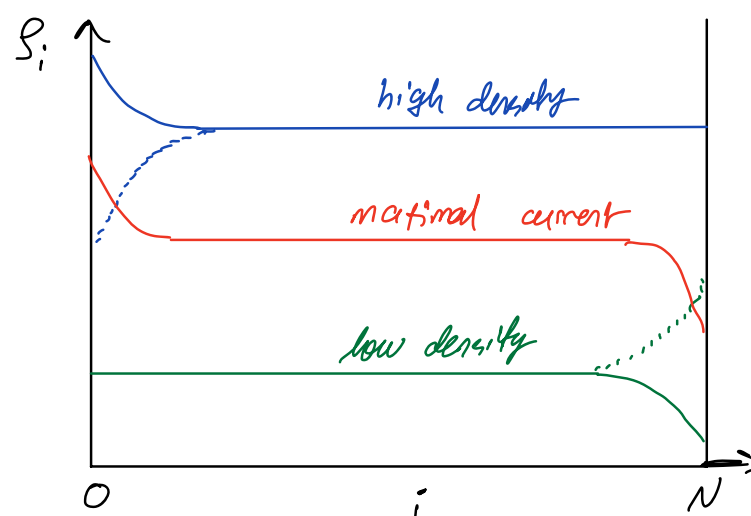
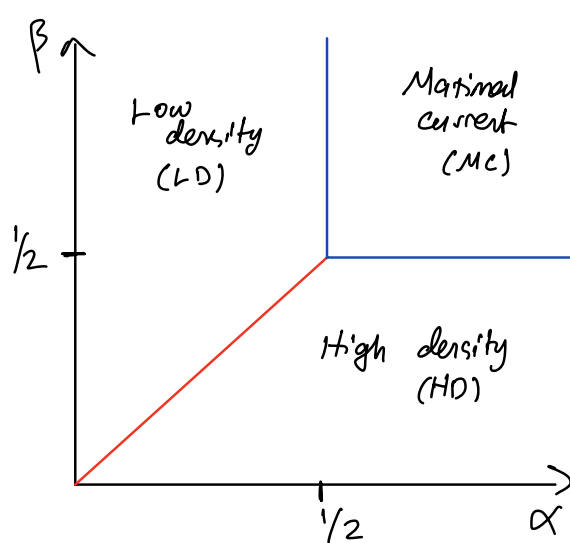
Introduced by Frank Spitzer in 1970, the TASEP is a paradigmatic model in the theory of transport phenomena and nonequilibrium statistical mechanics more broadly. It comprises particles on a 1-D lattice which hop rightward to vacant sites with rate  $p$ .



We will be looking at the case of open boundaries, where particles enter site  $i=1$  with rate  $\alpha$  if  $i=1$  is vacant, and particles are lost from site  $i=N$  at rate  $\beta$  if it is occupied. The case of periodic boundaries is less interesting and will be treated exactly in your problem set.

In the following, we work in time units where  $p=1$ . We let  $\vec{n} = (n_1, n_2, \dots, n_N)$  denote the vector of occupancy numbers  $n_i \in \{0, 1\}$ . The local density is defined as  $\rho_i \equiv \langle n_i \rangle$ .

This model turns out to have a rich phase diagram:



To understand this, we will derive a mean-field dynamics for  $\rho_i \equiv \langle n_i \rangle$

### Expectation values in Jump processes

Suppose you have some observable of your state  $f(\phi)$

What is the time evolution of  $\langle f(\phi) \rangle$ ?

$$\begin{aligned}
\partial_t \langle f(\varphi) \rangle &= \sum_{\varphi} f(\varphi) \partial_t P(\varphi, t) \\
&= \sum_{\varphi, \varphi'} f(\varphi) P(\varphi') \underbrace{\omega(\varphi' \rightarrow \varphi)}_{\text{interchange } \varphi \leftrightarrow \varphi'} - f(\varphi) P(\varphi) \omega(\varphi \rightarrow \varphi') \quad (\text{master eqn}). \\
&= \sum_{\varphi, \varphi'} (f(\varphi') - f(\varphi)) \omega(\varphi \rightarrow \varphi') P(\varphi) \\
&= \sum_{\varphi'} \langle (f(\varphi') - f(\varphi)) \omega(\varphi \rightarrow \varphi') \rangle
\end{aligned}$$

Note that we need only sum over those  $\varphi'$  which change  $f$ .

### Mean field theory of the TASEP

Applying this to occupancy numbers  $f(\vec{n}) = n_i$ , we need only count hops into and out of  $i$ :

$$\begin{aligned}
\partial_t \langle n_i \rangle &= \langle (+1) \cdot W(\text{hop from } i-1 \text{ to } i) \rangle + \langle (-1) \cdot W(\text{hop from } i \text{ to } i+1) \rangle \\
\partial_t \langle n_i \rangle &= \langle n_{i-1} (1 - n_i) \rangle - \langle n_i (1 - n_{i+1}) \rangle
\end{aligned}$$

We see that to obtain the dynamics of the one-point function  $\langle n_i \rangle$  requires knowledge of the two-point function  $\langle n_i n_j \rangle$ . If we were to write a dynamics for  $\langle n_i n_j \rangle$ , we would have a dependence on  $\langle n_i n_j n_k \rangle$ , and so on, leading to an infinite hierarchy of equations. This is a common theme in interacting many-body problems.

To make progress, we must truncate the hierarchy. A standard approximation is to truncate at the one-point level - a so called mean-field (MF) approximation:

$$\langle n_i n_j \rangle \approx \langle n_i \rangle \langle n_j \rangle$$

We then have

$$\partial_t \langle n_i \rangle = \langle n_{i-1} \rangle (1 - \langle n_i \rangle) - \langle n_i \rangle (1 - \langle n_{i+1} \rangle).$$

$$\Rightarrow \dot{S}_i = S_{i-1} (1 - S_i) - S_i (1 - S_{i+1}). \quad *$$

Since we are interested in long-wavelength behaviour, we will use a coarse-grained continuum description. Let  $a$  be the lattice spacing. Define a continuous variable:

$$x = i a \in (0, L), \text{ where } L \equiv Na.$$

and a field:  $S(x) \equiv S_{x/a}$

We then have  $S_i = S(x)$

$$S_{i\pm 1} = S(x \pm a) = S(x) \pm a \partial_x S(x) + \frac{a^2}{2} \partial_x^2 S(x) + \mathcal{O}(a^3)$$

Plugging into  $*$ , we have:

$$\begin{aligned} \partial_t S &= (S - a \partial_x S + \frac{a^2}{2} \partial_x^2 S)(1 - S) - S(1 - S - a \partial_x S - \frac{a^2}{2} \partial_x^2 S) \\ &= -a \partial_x S + 2a S \partial_x S + \frac{a^2}{2} \partial_x^2 S \end{aligned}$$

$$\Rightarrow \boxed{\partial_t S = -a \partial_x \left[ S(1-S) - \frac{a}{2} \partial_x S \right]}$$

$$= -\partial_x J, \quad \text{where } D = a^2/2$$

$$J(x) = a S(1-S) - D \partial_x S$$

We see this to take the form of a continuity equation, reflecting the local conservation of particle number in the bulk.

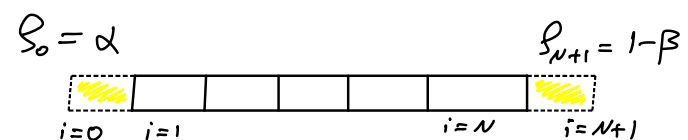
How about boundary conditions?

Notice that our open boundaries are equivalent to having reservoirs at either end with densities  $S_{i=0} = \alpha$  and  $S_{i=N+1} = 1 - \beta$

Thus:

$$S(0) = \alpha$$

$$S(L) = 1 - \beta$$



### Mean field Phase diagram

For simplicity, set  $a=1$  by rescaling units of space. The steady-state density profile satisfies:

$$\partial_x [S(1-S) - \frac{1}{2} \partial_x S] = 0$$

$$\Rightarrow S(1-S) - \frac{1}{2} \partial_x S = J_0, \quad \text{where } J_0 \text{ is undetermined.}$$

This equation can be solved by separation of variables:

$$\frac{dS}{dx} = 2S(1-S) - 2J_0 \quad **$$

$$\int_{S(0)=\alpha}^{S(x)} \frac{dS}{S(1-S) - J_0} = 2x. \quad \Rightarrow \text{solve for } S(x)$$

For  $J_0 < 1/4$ , one finds solutions of the form:

$$S(x) = \frac{1}{2} - \frac{1}{2} g \tanh\left[gx + \operatorname{arctanh}\left(\frac{1-2\alpha}{g}\right)\right], \quad \text{where } g \equiv \sqrt{1-4J_0}.$$

The value of  $g$  and the current  $J$  is then determined using  $S(L) = 1-\beta$ .

This is a transcendental equation which can be analyzed for  $L \gg 1$  - we won't do this here, but see Blythe and Evans 2007 J. Phys. A. 40 B333.

In summary, we find that the density profile is flat in the bulk

$$S(x) = S_0 \quad (\text{far from boundaries})$$

and that  $S_0$  can take three possible values, depending on the values of  $\alpha, \beta$ :

- 1)  $\alpha < \beta$ ,  $\alpha < 1/2$ :  $S_0 = S(0) = \alpha$  (low density)
- 2)  $\beta < \alpha$ ,  $\beta < 1/2$ :  $S_0 = S(L) = 1-\beta$  (high density)
- 3)  $\alpha > 1/2$ ,  $\beta > 1/2$ :  $S_0 = 1/2$  (maximal current)

The current must be constant in space in steady state  $J(x) = J_0 = S(1-S) - S'(x)/2$ .

In the bulk,  $S'(x) = 0$ , so that

$$J_0 = S_0(1-S_0) \begin{cases} \alpha(1-\alpha) & \text{(LD)} \\ \beta(1-\beta) & \text{(HD)} \\ 1/4 & \text{(MC)} \end{cases}$$

